🩺 End-to-End Melanoma Detection with Custom Attention & Grad-CAM Explainability

An end-to-end deep learning framework built with **PyTorch** for automated skin lesion classification and Explainable AI (XAI). This repository harmonizes multi-source benchmark datasets (**ISIC 2018** & **HAM10000**), evaluates 4 foundational baselines, and implements a custom **Channel Attention EfficientNet** model paired with **Grad-CAM** visual heatmaps.

📌 Features & Highlights

- **Data Harmonization Pipeline**: Automatically cleans and merges metadata from 11,527 dermoscopic images across ISIC 2018 and HAM10000 datasets into a standardized binary schema.
- **Stratified Dataset Splitting**: Strict 80/10/10 split ensuring zero data leakage across train (`9,221`), validation (`1,153`), and test (`1,153`) sets.
- **Multi-Baseline Benchmarking**: Systematic comparison of `EfficientNet-B0`, `ResNet-50`, `DenseNet-121`, and `MobileNetV3`.
- **Custom Channel Attention Network**: Integrated Squeeze-and-Excitation attention block to re-weight feature maps toward critical diagnostic lesion features.
- **Explainable AI (Grad-CAM)**: Built-in backward hooks to visualize network focus areas over lesion boundaries.

📊 Experimental Results & Benchmark

All models were evaluated on the harmonized validation split using Area Under the ROC Curve (AUC-ROC):

Model Architecture	Parameters	Val AUC-ROC	Key Characteristic
EfficientNet-B0 (Baseline)	5.3M	**0.9166**	Highly parameter efficient
Custom Attention Model (Ours)	5.9M	**0.9135**	Dynamically recalibrates feature maps for diagnostic focus
MobileNetV3 Large	5.4M	**0.9024**	Lightweight edge deployment candidate

Model Architecture	Parameters	Val AUC-ROC	Key Characteristic
DenseNet-121	8.0M	**0.9013**	Dense feature reuse across layers
ResNet-50	25.6M	**0.9001**	Deep residual baseline

🏗️ Model Architecture (Channel Attention)

The custom model embeds a Channel Attention module prior to adaptive average pooling:

```
None
Input (224x224x3) ⟶ EfficientNet-B0 Features (1280x7x7) ⟶ Channel Attention
⟶ Pooling ⟶ Classifier ⟶ Output (Logit)
                                                                    |
                                                          [AvgPool + MaxPool -> FC ->
Sigmoid]
```

🚀 Getting Started

Prerequisites

```
Shell
pip install torch torchvision pandas numpy opencv-python matplotlib scikit-learn
pillow
```

Running in Kaggle / Local Notebook

1. Clone or copy the notebook into your environment.
2. Enable GPU acceleration (`CUDA`).
3. Ensure datasets (`isic-2018-jpg-256x256-resized` and `skin-cancer-mnist-ham10000`) are mounted in the input directory.
4. Execute cells sequentially.

👁️ Visual Explainability (Grad-CAM)

The model includes automated Grad-CAM visual extraction:

```python
# Generate heatmap over final feature layer
target_layer = attention_model.features[-1]
grad_cam = GradCAM(attention_model, target_layer)
```

📜 Citation

If you use this project or methodology in your work, please cite:

```bibtex
@article{melanoma_attention_2026,
  title={Harmonized Multi-Dataset Melanoma Classification via Channel Attention Networks and Explainable Visual Artifact Analysis},
  author={Machine Learning Research Team},
  journal={GitHub Repository},
  year={2026}
}
```